

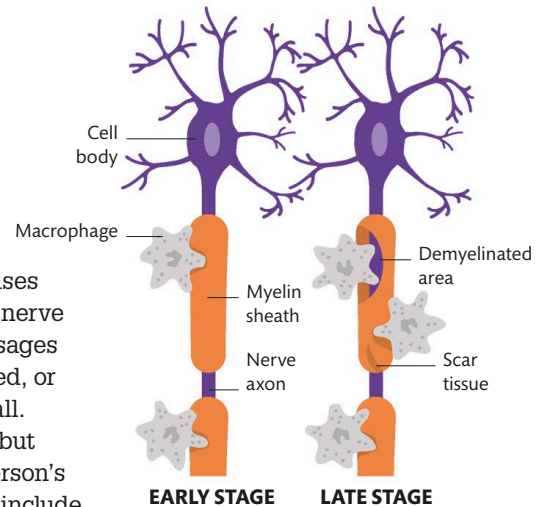
Multiple sclerosis

Multiple sclerosis, or MS, is a condition that affects both the brain and the spinal cord. It is believed to be caused when the body's immune system mistakenly damages protective nerve sheaths.

Myelin cells, made of proteins and fats, surround neurons in the central nervous system, enabling messages to travel quickly and smoothly between the brain and the rest of the body. When MS develops, the immune system, which normally fights infection and inflammation, seems to mistake myelin for a foreign body and attacks it with macrophage cells, damaging it and stripping it away. The scars, or plaques, this action

leaves behind disrupt impulses normally transmitted along nerve fibers or axons. Neural messages slow down, become distorted, or are simply not delivered at all.

MS may occur at any age but is usually diagnosed in a person's 20s or 30s. Early symptoms include dizziness, vision changes, and muscle weakness. In later stages, speech, mobility, and cognition may be affected. The progressive form of the disease results in disability.



Macrophage numbers and MS stages

When MS begins, macrophage cells remove damaged tissue but also help repair it. In later stages, however, their numbers increase and actually accelerate myelin loss, increasing the severity of symptoms.

Motor neuron disease

Motor neuron disease, or MND, is an umbrella term used to describe a group of conditions that affect motor neurons—the nerves in the brain and spinal cord that tell all the muscles in the body what to do.

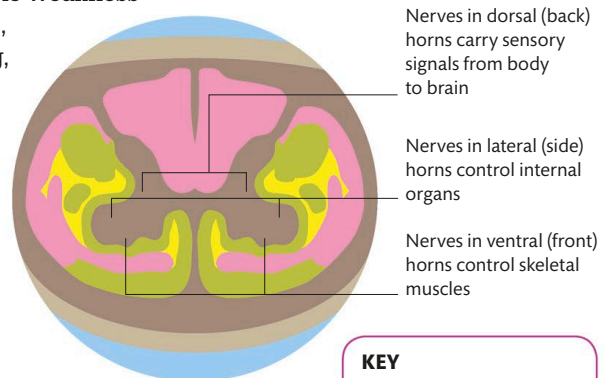
Genetic, environmental, and lifestyle factors are thought to contribute to the development of MND. Exposure to heavy metals or agricultural chemicals, an electrical or mechanical trauma, military service, or excessive exercise have all been investigated as possible causes, with conflicting results.

Some types of MND, however, do have a genetic basis. Progressive bulbar atrophy, also known as Kennedy disease, results from a mutated gene and affects mainly men. Kennedy disease specifically damages the bulb-shaped lower brain stem, where neurons that control muscles in the face and throat are found.

Whatever their cause, most forms of MND cause symptoms that include general muscle weakness and wasting, cramps, difficulty swallowing, a progressive loss of speech, and limb weakness. Diagnosis includes MRI scans, muscle biopsies, and blood and urine tests.

Although there is currently no cure for MND, symptoms, can be managed to give sufferers the best possible quality of life.

PHYSICIST
STEPHEN HAWKING LIVED FOR
55 YEARS AFTER BEING
DIAGNOSED WITH MND



Spinal cord bundles

Different forms of MND involve different tracts of neurons, located in the dorsal, lateral, and ventral horns of the spinal cord.